

Project Interim Report
On
Geometric Shape Area Calculator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

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PROJECT REPORT CONTENT

Title

Design and Implementation of a Geometric Shape Area Calculator using Abstraction in Object-Oriented Programming

Abstract

This project focuses on developing a Geometric Shape Area Calculator using Java and Object-Oriented Programming concepts, particularly abstraction and polymorphism. The system is designed to calculate the area of different geometric shapes such as Circle and Square using a common interface.

In mathematics, calculating areas is a fundamental operation, and this project provides a structured and reusable approach to solving it programmatically. The Shape interface defines a common method `calculateArea()`, which is implemented differently by each shape class based on its formula.

The project demonstrates abstraction by hiding implementation details and exposing only essential functionality. It also showcases polymorphism, where a single

interface reference is used to call different implementations at runtime. This design makes the system flexible and extensible, allowing new shapes to be added easily without modifying existing code.

Introduction

The Geometric Shape Area Calculator is a Java-based application that calculates the area of various shapes using Object-Oriented Programming principles. It uses interfaces to define a common structure and ensures code reusability and modular design.

Objectives

To implement abstraction using interfaces

To calculate areas of geometric shapes

To demonstrate polymorphism using interface references

To create reusable and modular code

To design an extensible system for adding new shapes

This project can be used for educational purposes and basic geometry calculations. It can be extended by:

Adding more shapes (Triangle, Rectangle, etc.)

Creating a graphical user interface

Integrating with real-world applications

System Requirements

Hardware:

Computer with 4GB RAM

Software:

Java JDK

IDE (VS Code / Eclipse)

Methodology

The system works as follows:

Define a Shape interface with a method calculateArea()

Create classes like Circle and Square implementing the interface

Each class provides its own formula for area calculation

Use Shape reference to create objects (polymorphism)

Call calculateArea() method dynamically

Display the calculated area

Flowchart

Start



Create Shape Objects



Assign values (radius/side)



Call calculateArea()



Display Area



End

Class Diagram

Write in box format:

Shape (Interface)

+ calculateArea()



Circle

Square

- radius

- side

+ calculateArea()

Implementation

Shape Interface

Defines a method `calculateArea()` that must be implemented by all shapes.

Circle Class

Stores radius

Calculates area using formula $\pi \times r^2$

Square Class

Stores side length

Calculates area using `side × side`

Main Class

Uses Shape reference

Demonstrates polymorphism

Calls methods and prints output

Sample Output

Area of Circle: 78.5

Area of Square: 16.0

Gantt Chart

TaskDuration Timeline

Planning 1 Day Day 1

Design (Interface & Class) 1 Day Day 2

Coding 2 Days Day 3-4

Testing 1 Day Day 5

Report Writing 1 Day Day 6

Conclusion

The Geometric Shape Area Calculator successfully demonstrates the use of abstraction and polymorphism in Java. By using an interface, the project ensures that all shapes follow a common structure while allowing flexibility in implementation.

Each class provides its own logic for calculating area, which highlights the concept of method overriding. The use of interface references to call different implementations at runtime demonstrates polymorphism effectively.

Overall, the project provides a clean, modular, and extensible design that can be easily expanded to include more shapes and functionalities in the future.

Future Enhancements

Add more shapes (Triangle, Rectangle)

GUI using Java Swing

User input system

Store results in file/database

References

[1] H. Schildt, Java: The Complete Reference

[2] Oracle Java Documentation

JAVA CODE (FINAL)

```
interface Shape {  
    double calculateArea();  
}
```

// Circle Class

```
class Circle implements Shape {  
    private double radius;  
  
    public Circle(double radius) {  
        this.radius = radius;  
    }  
  
    public double calculateArea() {  
        return Math.PI * radius * radius;  
    }  
}
```

// Square Class

```
class Square implements Shape {
```

```
    private double side;
```

```
    public Square(double side) {
```

```
        this.side = side;
```

```
    }
```

```
    public double calculateArea() {
```

```
        return side * side;
```

```
    }
```

```
}
```

```
// Main Class
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Shape s1 = new Circle(5);
```

```
        Shape s2 = new Square(4);
```

```
        System.out.println("Area of Circle: " +  
s1.calculateArea());
```

```
        System.out.println("Area of Square: " +  
s2.calculateArea());
```

```
    }
```

```
}
```